University of Cape Town
Department of Mathematics and Applied Mathematics

2IA Class Test 2
17th October, 2022
Time: 1,5 hours
Full marks: 100

- **Read the instructions carefully.**
- **You have 1,5hrs to complete the test.**
- **Make sure that no pages or questions are missing.**
- **Show all your work or reasoning. Answers without work or reasoning shown will receive no mark.**

Surname: ______________________

First name: ______________________

Student number: ______________________

Signature: ______________________

Do not write below this line:

Question:	1	2	3	4	Total
Marks:	21	33	12	34	100
Score:					

1. (a) $\boxed{\text{3 marks}}$ Give an example of an even permutation.

   (b) $\boxed{\text{3 marks}}$ Give an example of an odd permutation.

   (c) $\boxed{\text{10 marks}}$ Prove that the product of any two (distinct) transpositions of S_n is a product of 3-cycles.

   (d) $\boxed{\text{5 marks}}$ Using (c), show that any even permutation can be written as a product of 3-cycles.

Solution: (a) We proved in class that the identity permutation is even. Otherwise, any 3-cycle is even since it is the product of two transpositions. For instance,

$$(1 \quad 2 \quad 3) = (1 \quad 3)(1 \quad 2)$$

(b) Any transposition is an odd permutation, for example, the transposition $(1 \quad 2)$.

(c) Let α and β be two distinct transpositions in S_n. We distinguish two cases:

(1) α and β move one common element.

Then α and β are of the form $\alpha = (a \quad b)$, $\beta = (a \quad c)$, for some $a, b, c \in X_n$ with $a \notin \{b, c\}$ and $b \neq c$. In this case we can rewrite $\alpha\beta$ as a product of only one factor 3-cycle; that is, $\alpha\beta$ is a 3-cycle. More precisely, we claim:

$$\alpha\beta = (a \quad b)(a \quad c) \stackrel{?}{=} (a \quad c \quad b)$$

Let $\gamma = (a \quad c \quad b)$. To show that $\alpha\beta = \gamma$ we need to prove that $(\alpha\beta)(k) = \gamma(k)$ for all $k \in X_n$. This clearly holds for $k \notin \{a, b, c\}$, since in that case, we have that $(\alpha\beta)(k) = \gamma(k) = k$. Now, for $k = a$ we have that

$$(\alpha\beta)(a) = \alpha(\beta(a)) = \alpha(c) = c,$$
$$(\gamma)(a) = c.$$

For $k = b$ we have that

$$(\alpha\beta)(b) = \alpha(\beta(b)) = \alpha(b) = a,$$
$$(\gamma)(b) = a.$$

Lastly, for $k = c$ we obtain that

$$(\alpha\beta)(c) = \alpha(\beta(c)) = \alpha(a) = b,$$
$$(\gamma)(c) = b.$$

This shows that $\alpha\beta = \gamma$, and we are done in this case.

(2) α and β are disjoint.

In such a case, α and β are of the form $\alpha = (a \quad b)$, $\beta = (c \quad d)$, for some $a, b, c, d \in X_n$ with $a \neq b$, $c \neq d$ and $\{a, b\} \cap \{c, d\} = \emptyset$. We claim that $\alpha\beta$ is the following product of 3-cycles:

$$\alpha\beta = (a \quad b)(c \quad d) \stackrel{?}{=} (a \quad c \quad d)(a \quad b \quad d)$$

Let $\delta_1 = (a \quad c \quad d)$ and $\delta_2 = (a \quad b \quad d)$ and let us prove that $(\alpha\beta)(k) = (\delta_1\delta_2)(k)$ for all $k \in X_n$. For $k \notin \{a, b, c, d\}$ we have that $(\alpha\beta)(k) = (\delta_1\delta_2)(k) = k$; for $k = a$ we have that

$$(\alpha\beta)(a) = \alpha(\beta(a)) = \alpha(a) = b,$$
$$(\delta_1\delta_2)(a) = \delta_1(\delta_2(a)) = \delta_1(b) = b,$$

proving that $(\alpha\beta)(a) = (\delta_1\delta_2)(a)$; for $k = b$ we have that

$$(\alpha\beta)(b) = \alpha(\beta(b)) = \alpha(b) = a,$$
$$(\delta_1\delta_2)(b) = \delta_1(\delta_2(b)) = \delta_1(d) = a,$$

and $(\alpha\beta)(b) = (\delta_1\delta_2)(b)$; for $k = c$ we have that

$$(\alpha\beta)(c) = \alpha(\beta(c)) = \alpha(d) = d,$$
$$(\delta_1\delta_2)(c) = \delta_1(\delta_2(c)) = \delta_1(c) = d,$$

which shows that $(\alpha\beta)(c) = (\delta_1\delta_2)(c)$; lastly for $k = d$ we have that

$$(\alpha\beta)(d) = \alpha(\beta(d)) = \alpha(c) = c,$$
$$(\delta_1\delta_2)(d) = \delta_1(\delta_2(d)) = \delta_1(a) = c,$$

and $(\alpha\beta)(d) = (\delta_1\delta_2)(d)$.

In any case, we have proved that $\alpha\beta$ is a product of 3-cycles.

(d) Let σ be an even permutation. This means that σ is a product of an even number of transpositions. Grouping the factor transpositions of σ in pairs and applying (d) to each of these pairs we can express σ as a product of 3-cycles, as requested.

2. In each case determine whether the statement is true or false. **You will get zero marks if you do not provide an argument supporting your answer.**

   (a) $\boxed{\text{5 marks}}$ The set of all permutations S_3 is an abelian group.

   (b) $\boxed{\text{5 marks}}$ The permutation $\sigma = (1\ \ 6\ \ 7\ \ 3\ \ 5)(4\ \ 7\ \ 6\ \ 3)(2\ \ 1\ \ 5)$ of S_7 fixes 1 and 7.

   (c) $\boxed{\text{6 marks}}$ Every abelian group is cyclic.

   (d) $\boxed{\text{6 marks}}$ If G and H are cyclic groups, then its Cartesian product $G \times H$ is also a cyclic group.

   (e) $\boxed{\text{5 marks}}$ Every nontrivial group has at least two subgroups.

   (f) $\boxed{\text{6 marks}}$ $\bar{2}$ is a generator of the (additive) cyclic group $\mathbb{Z}_{15}$.

Solution: (a) **False:** The transpositions $\alpha = (1\ \ 2)$ and $\beta = (1\ \ 3)$ are in S_n and they do not commute. In fact:

$$\begin{aligned}\alpha\beta &= (1\ \ 2)(1\ \ 3) = (1\ \ 3\ \ 2)\\ \beta\alpha &= (1\ \ 3)(1\ \ 2) = (1\ \ 2\ \ 3)\end{aligned}$$

(b) **True:** Let $\sigma_1 = (2\ \ 1\ \ 5)$, $\sigma_2 = (4\ \ 7\ \ 6\ \ 3)$ and $\sigma_3 = (1\ \ 6\ \ 7\ \ 3\ \ 5)$. Notice that $\sigma = \sigma_3\sigma_2\sigma_1$, and so

$$\begin{aligned}&1 \xmapsto{\sigma_1} 5 \xmapsto{\sigma_2} 5 \xmapsto{\sigma_3} 1,\\ &7 \xmapsto{\sigma_1} 7 \xmapsto{\sigma_2} 6 \xmapsto{\sigma_3} 7,\end{aligned}$$

which show that $\sigma(1) = 1$ and $\sigma(7) = 7$.

(c) **False:** As seen in class the group $\mathbb{Z}_2 \times \mathbb{Z}_2$ is not cyclic (all its elements have order 2) and it is abelian.

(d) **False:** We can consider the same example that in (c). Take $G = H = \mathbb{Z}_2$, then as said in (c), the group $\mathbb{Z}_2 \times \mathbb{Z}_2$ is not cyclic.

(e) **True:** If G denotes any nontrivial group, then $\{1\}$ and G are two distinct subgroups of G.

(f) **True:** We proved in class that the generators of $\mathbb{Z}_{15}$ are of the form $\bar{a}$, where $\gcd(a, 15) = 1$.

3. $\boxed{\text{12 marks}}$ Let G be a group. Prove that G is abelian if and only if $(ab)^2 = a^2b^2$ for all $a, b \in G$.

Solution: Suppose first that G is abelian and take a and b in G. Then $ab = ba$ and

$$(ab)^2 = (ab)(ab) = a(ba)b = a(ab)b = (aa)(bb) = a^2b^2.$$

Conversely, assume that $(ab)^2 = a^2b^2$ for all $a, b \in G$, and let us show that G is abelian. Take $g, h \in G$ and let us show that $gh = hg$. Using the hypothesis, we have that $(gh)^2 = g^2h^2$; on the other hand, we have that $(gh)^2 = (gh)(gh) = ghgh$. From here we obtain that

$$\begin{aligned} g^2h^2 = ghgh &\Rightarrow g^{-1}(g^2h^2)h^{-1} = g^{-1}(ghgh)h^{-1} \\ &\Rightarrow (g^{-1}g)(gh)(hh^{-1}) = (g^{-1}g)hg(hh^{-1}) \Rightarrow gh = hg, \end{aligned}$$

proving that G is abelian.

4. (a) $\boxed{\text{4 marks}}$ Complete the missing parts (marked with a $\star$) in the statement of the **Fundamental Theorem of Finite Cyclic Groups**.

> <u>FUNDAMENTAL THEOREM OF FINITE CYCLIC GROUPS.</u> Let $G = \langle g \rangle$ be a cyclic group of order n.
> (1) If H is a subgroup of G, then $H = \langle g^d \rangle$ for some $(\star)$.
> (2) If H is any subgroup of G with order $|H| = k$, then $(\star)$.
> (3) If $k|n$, then $\langle g^{n/k} \rangle$ is $(\star)$ of G of order $(\star)$.

(b) Consider the following incomplete proof of (1) in the **Fundamental Theorem of Finite Cyclic Groups**:

> The result is clear for $|H| = 1$. Otherwise, we know that H is cyclic, and so $H = \langle g^m \rangle$. Let $d =$ (A). Then $d|n$ and we only need to show that $H = \langle g^d \rangle$. From $d|m$, we have that $m = qd$ for some $q \in \mathbb{Z}$. Thus (B) and therefore $H \subseteq \langle g^d \rangle$. On the other hand, by (C) we have that $d = xm + yn$, where $x, y \in \mathbb{Z}$, and so
>
> (D),
>
> which implies that $\langle g^d \rangle \subseteq H$, and (1) follows.

With reference to this partial proof:

i. $\boxed{\text{3 marks}}$ Complete the sentence at (A)

ii. $\boxed{\text{5 marks}}$ Fill in the necessary steps at (B) to conclude that $H \subseteq \langle g^d \rangle$.

iii. $\boxed{\text{2 marks}}$ Name the result you are using in (C).

iv. $\boxed{\text{5 marks}}$ Fill in the necessary steps at (D) to conclude that $\langle g^d \rangle \subseteq H$.

(c) Consider the additive cyclic group $\mathbb{Z}_{18}$.

i. $\boxed{\text{7 marks}}$ Find all the subgroups of $\mathbb{Z}_{18}$.

ii. $\boxed{\text{5 marks}}$ For each of the subgroups of $\mathbb{Z}_{18}$, write down all its elements.

iii. $\boxed{\text{3 marks}}$ Draw the lattice diagram of $\mathbb{Z}_{18}$.

Solution:

(a) <u>Fundamental Theorem of Finite Cyclic Groups.</u> Let $G = \langle g \rangle$ be a cyclic group of order n.

(1) If H is a subgroup of G, then $H = \langle g^d \rangle$ for some $d|n$.

(2) If H is any subgroup of G with order $|H| = k$, then $k|n$.

(3) If $k|n$, then $\langle g^{n/k} \rangle$ is the unique subgroup of G of order k.

(b) i. $d = \gcd(m, n)$.

ii. $g^m = g^{qd} = (g^d)^q \in \langle g^d \rangle$.

iii. Bézout's Lemma.

iv. $g^d = g^{xm+yn} = (g^m)^x (g^n)^y = (g^m)^x \in H$.

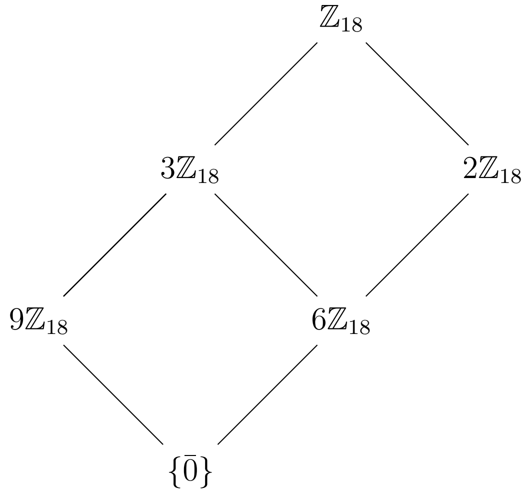
(c) Since the divisors of 18 are 1, 2, 3, 6, 9, 18, by the Fundamental Theorem of Finite Cyclic Groups we can conclude that $\mathbb{Z}_{18}$ has exactly 6 subgroups of orders 1, 2, 3, 6, 9 and 18. It is clear that the subgroups of orders 1 and 18 are $\{\bar{0}\}$ and $\mathbb{Z}_{18}$, respectively. On the other hand, we have that

$$2 = |d\bar{1}| = \frac{18}{2} \Rightarrow d = 9; \text{ the subgroup of order 2 is } \langle \bar{9} \rangle = 9\mathbb{Z}_{18};$$
$$3 = |d\bar{1}| = \frac{18}{3} \Rightarrow d = 6; \text{ the subgroup of order 3 is } \langle \bar{6} \rangle = 6\mathbb{Z}_{18};$$
$$6 = |d\bar{1}| = \frac{18}{6} \Rightarrow d = 3; \text{ the subgroup of order 6 is } \langle \bar{3} \rangle = 3\mathbb{Z}_{18};$$
$$9 = |d\bar{1}| = \frac{18}{9} \Rightarrow d = 2; \text{ the subgroup of order 9 is } \langle \bar{2} \rangle = 2\mathbb{Z}_{18};$$

Moreover:

$$9\mathbb{Z}_{18} = \{\bar{0}, \bar{9}\};$$
$$6\mathbb{Z}_{18} = \{\bar{0}, \bar{6}, \overline{12}\};$$
$$3\mathbb{Z}_{18} = \{\bar{0}, \bar{3}, \bar{6}, \bar{9}, \overline{12}, \overline{15}\};$$
$$2\mathbb{Z}_{18} = \{\bar{0}, \bar{2}, \bar{4}, \bar{6}, \bar{8}, \overline{10}, \overline{12}, \overline{14}, \overline{16}\};$$

(b) By (a) the lattice diagram of $\mathbb{Z}_{18}$ is as follows:


$\mathbb{Z}_{18}$
$3\mathbb{Z}_{18}$
$2\mathbb{Z}_{18}$
$9\mathbb{Z}_{18}$
$6\mathbb{Z}_{18}$
$\{\bar{0}\}$